Neurosciences (research)

Vrije Universiteit Amsterdam - Faculty of Science - Neurosciences (research) - 2025-2026

Brain disorders, in particular neurodegenerative diseases and mental illnesses, are among the most prevalent and debilitating diseases of our time. They are typically caused by the interplay of environmental factors and genetic variation in multiple genes, which is currently being mapped by major international initiatives. Future brain research should therefore focus on integrative projects as a next step in characterizing this complex environmental/genetic interplay and in revealing how it translates into brain function and/or disease. To succeed, we need a new generation of neuroscientists capable of integrating information from genes and proteins to synapses, and from networks up to complex brain function and dysfunction with relevant neurocomputational skills. Are you ready for the challenge?

The first year consists of compulsory courses to give you a thorough grounding in relevant disciplines, followed by a five month internship. You will have a wide range of placement opportunities to choose from within the Amsterdam Neuroscience community and affiliated institutes.

The second year consists of advanced coursework to expand your knowledge of key research areas. These elective courses are subdivided in two study paths, namely 'Clinical and translational neurosciences' and 'Fundamental neurosciences'. Please note that some elective courses may have a capacity restriction. Alternative courses are offered in parallel within the programme (possibly as part of another study path). During the last six months of the programme, you will complete your second internship in the Netherlands or abroad.

If you are a prospective student or would like to know more about this program, please go to https://masters.vu.nl/en/programmes/neurosciences/

Info

Level	Master
Language	English
Duration	2 years
Type	Full Time
Credits	120 EC
Faculty	Faculty of Science

Teaching and examination regulation 2025-2026
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Neurosciences year 1

Description

The first year consists of compulsory courses, to give you a thorough grounding in relevant disciplines. These courses are followed by a 5-month internship placement within the Amsterdam Neuroscience community or at an affiliated institute.

Courses

Course Name	Period	Credits	Code
Career and Academic Skills Portfolio year 1 Neurosciences	Ac. Year (September)	0EC	AM_1250
Internship Neurosciences I	Ac. Year (September)	27EC	AM_471108
From Molecule to Mind	P1	6EC	AM_1275
Statistics in Neurosciences	P1	6EC	AM_1276
Clinical neurosciences	P2	6EC	AM_1005
Genetics in Neuroscience	P2	6EC	AM_1214
Neurogenomics	P3	6EC	AM_1007
Academic Writing: Neurosciences	P4	3EC	AM_1123

Neurosciences year 2

Description

The first semester of the second year contains electives, to expand your knowledge of certain areas. We have constructed two study paths, namely Clinical and translational neurosciences, and Fundamental neurosciences, which give you a perfect start as a researcher in a particular field of neurosciences.

Programme Sections

- Neuroscience year 2 compulsory courses
- Neuroscience year 2 suggested electives Fundamental Neurosciences
- Neuroscience year 2 suggested electives Clinical and Translational Neurosciences

Neuroscience year 2 compulsory courses

Courses

Course Name	Period	Credits	Code
Career and Academic Skills and Portfolio year 2 Neurosciences	Ac. Year (September)	3EC	AM_1268
Final Year Project Neurosciences	Ac. Year (September)	30EC	AM_1269
Neurophilosophy and Ethics	P3	3EC	AM_1018

Neuroscience year 2 suggested electives Fundamental Neurosciences

Description

The Fundamental neuroscience study path aims to provide students with fundamental knowledge and hands-on experience with state of the art neuroscience technology to prepare for a PhD in fundamental neuroscience. You will study the effect of genes and gene mutations in systems of increasing complexity, from single neurons to networks and whole organisms using the latest technologies in molecular neurobiology, stemcell and human neuron preparations, and live cell imaging.

Courses

Course Name	Period	Credits	Code
Advanced Neurogenomics	P1	6EC	AM_470717

Course Name	Period	Credits	Code
Cellular Models for Brain Disorders	P1	6EC	AM_1218
Live Cell Imaging	P1	6EC	AM_470726
Neuron-Glia Interactions	P1	6EC	AM_1217
Behavioral Neurosciences	P2	6EC	AM_1299
Developmental Neurobiology of the Vertebrate Brain	P2	6EC	AM_470713
Network Neuroscience	P2	6EC	AM_1316
Systems Neuroscience	P2	6EC	AM_470712

Neuroscience year 2 suggested electives Clinical and Translational Neurosciences

Description

The Clinical and Translational Neurosciences study path makes use of all fundamental knowledge and understanding gathered in the first year in order to evaluate, improve and design clinical research in neuroscience, neurology and psychiatry. Translating from fundamental to clinical, you will link pathophysiological properties to symptomatology and treatment outcomes. Translating from clinical to fundamental, you will learn how to navigate the clinical realm of brain disease and disorders, define knowledge gaps, and create experiments that might fill those gaps.

Courses

Course Name	Period	Credits	Code
Advanced Clinical Neurosciences	P1	6EC	AM_1014
Functional Brain Imaging	P1	6EC	AM_470715
Experimental and Clinical Neuroendocrinology	P2	6EC	AM_470700
Neuro- and Psychopharmacology	P2	6EC	AM_470718
Rhythms of the Brain	P2	6EC	AM_1003

Courses

Academic Writing: Neurosciences

Course Code	AM_1123
Credits	3
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. G.A. Dreschler
Examiner	dr. G.A. Dreschler
Teaching Staff	
Teaching method(s)	Study Group

Course Objective

On successful completion of this course you will be a more confident writer in English in terms of producing an academic text which is coherent and linguistically correct. You will have insight into the most important stylistic aspects of academic texts in your discipline. You will also have insight into how specific linguistic structures contribute to text coherence, and will have the ability to identify and improve aspects of coherence in your own academic texts. Finally, you will have clear insight into your strengths and weaknesses of your academic English writing skills, and knowledge of how to further develop your strengths and reduce your weaknesses.

Course Content

The course will start off by ensuring that you are up to speed in terms of the macrostructural and discursive features of academic texts in general and the specific kind of text you are writing in the content of the Neurosciences programme. Centre stage is then given to aspects of coherent text and to linguistic devices that English uses to express complex ideas: we introduce basic principles, give you practice in analysing and reformulating defective text segments, and give you a checklist that you can use when editing your own text. The

idea underlying this approach is that writing essentially involves rewriting and rewriting.

The treatment of linguistic accuracy and academic style takes a less central place in the group sessions of the course, as prior knowledge of these is assumed, but the detailed feedback given on your texts allows you to work on any individual problems you may have, using the ELS-Online feedback site.

Additional Information Teaching Methods

Six two-hour seminars. The sessions are on campus and have required attendance (5 out of 6 sessions). If you miss a second session, you can finish the course if you make an alternative assignment. Once the schedule is published on rooster.vu.nl, make sure you make arrangements with your internship supervisors to attend these sessions.

Method of Assessment

The final product is the introduction to your internship report, approx. 1000 words in length and composed following agreed structural requirements. Your text will receive a pass/fail assessment. If you fail, you will be given feedback to help you prepare a reworked version.

Literature

There is no obligatory reading for this course, but for language support you can study the feedback categories in ELS-Online.

Additional Information

The materials for this course will be made available on the course's Canvas site. You will also be asked to submit any mid-course assignments plus your final assignment via Canvas. Please organize your Canvas account so that you receive course announcements additionally via your e-mail.

Advanced Clinical Neurosciences

Course Code	AM_1014
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. L. Douw
Examiner	dr. L. Douw
Teaching Staff	dr. L. Douw, dr. C. Vriend
Teaching method(s)	Seminar, Study Group, Lecture

Course Objective

The overall aim of the Advanced Clinical Neuroscience course is to learn how to design, perform and critically evaluate clinical neuroscience. Specific learning goals are as follows:

1. The student is able to design and interpret clinical neuroscientific research

- The student is able to list the most important clinical features of the five diseases/disorders that are central to this course (Alzheimer's disease, Parkinson's disease, multiple sclerosis, brain tumors, selected psychiatric disorders)
- The student is able to identify relevant diagnostic and treatment problems within these diseases/disorders
- The student is able to construct hypotheses directed at solving these diagnostic problems
- The student can design a research study about a clinical neuroscientific topic

2. The student can perform clinical neuroscientific research

- The student can recite techniques and methods used within neuroscience
- The student is able to apply and administer these methods within the five main diseases/disorders
- The student can argue what the correct statistics are to test his/her hypotheses

3. The student is able to critically evaluate clinical neuroscientific research

- The student is able to formulate pros and cons of a research proposal

- The student is able to apply their knowledge to criticize the scientific quality of clinical neuroscientific papers

Course Content

This course will involve (patient) lectures and workshops to familiarize students further with clinical neuroscience. They are expected to design a neuroscientific study of their own interest. They will integrate the current scientific state of the art with clinical patient presentation in a written case report. They will also gain experience in the process of peer review.

Additional Information Teaching Methods

Lectures (15 contact hours)

For each of the five main diseases/disorders, a clinical lecture will focus on a stringent/current problem that needs solving or will feature a patient. We assume basic knowledge on each diseases/disorders, so that we can discuss the problem in depth. Every clinical lecture is paired with a research lecture, in which an (innovative) approach towards solving a (perhaps not the same) problem.

Clinical neuroscience methods workshops (10 contact hours)

In order to further provide students with the tools necessary to conduct clinical neuroscience, a series of hands-on workshops will be taught. Topics include brain stimulation, scientific writing and presenting, epidemiology and trial design, peer-review etcetera.

Live attendence of all these lectures and workshops is mandatory to pass the course.

Method of Assessment

- Peer review assignment (20% of final grade)
- Written case report (50% of final grade)
- Study preregistration (30% of final grade)

NB all components have to be awarded a sufficient grade (5.5 or higher) in order to receive a sufficient final grade.

Entry Requirements

Clinical Neuroscience (470757)

Literature

A number of recent relevant reviews, peer-reviewed articles and chapters will be posted on Canvas.

Additional Information Target Audience

2nd year VU MSc Neurosciences students. Students may participate if they have passed the examination Clinical Neurosciences (AM_470757). NB: students who did not pass Clinical Neuroscience (470757) should contact the course coordinators before enrolling.

Additional Information

There is a maximum of 30 students. For further information and application, please contact the course coordinators: dr. Linda Douw or dr. Chris Vriend.

Custom Course Registration

Registration via VUnet, please inform us (l.douw@amsterdamumc.nl) if you switch after signing up through VUnet.

Advanced Neurogenomics

Course Code	AM_470717
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. M. Verhage

Examiner	prof. dr. M. Verhage
Teaching Staff	prof. dr. M. Verhage
Teaching method(s)	Practical

Course Objective

This course aims to introduce modern techniques in the integrated Neurosciences via hands-on experience. The course illustrates experimental approaches to seek causal relationship between molecular events, such as changes in gene-expression, gene-mutations, protein-protein interactions, and effects of such events on the. functioning of cells, networks and the whole organism. The use of these techniques is embedded in short research projects and typically covers a number of steps: first clone a construct (a mutant protein, an over-expression construct), than culture and tranfect a cell with this construct and finally analyze a functional consequences of this modification using functional assays. Alternatively, research projects may also start with the intact brain and then work down to the molecular networks of proteins and genes, for instance using proteomics and mRNA expression profiling. Although centered around genomics, many functional assays are offered, from electronmicroscopy and protein chemistry to life cell imaging, electrophysiology and behavioural analyses. Various tools and technologies will be explained in supporting lectures and used in order to answer questions related to the function of genes in the nervous system and how they contribute to higher order processes important for functioning of circuitry or behavior.

End terms:

- Profound insight and experience with neurogenomic techniques that are used during the course
- Adequate design, execution, and interpretation of neurogenomics experiments and conceptualisation of new working models (empirical cycle)
- Adequate oral presentation (2 oral presentations)

Course Content

This is a practical course. You will be executing a small research project aimed at illustrating today's research into gene-function relations and finding causal relationships between molecular events and functional consequences at the cellular/network level. The experiments involve gene expression analysis (transcripts, proteins), and genotypic and phenotypic analysis (genotyping, cell biology, behavior). Theoretical underpinning of concepts and methods will be achieved by using examples from recent literature.

Additional Information Teaching Methods

Lectures, discussion of papers, demonstrations, practicals
16x8h practicals, presentations and discussion
4x8h self study

Method of Assessment

Your final score for the course will depend on three items:

1. Pitch research question (15%)
2. Project practical (60%)
3. Presentation results and conclusion (25%)

Entry Requirements

Successful completion of the course Neurogenomics (1st year Master Neurosciences course; AM_1007) is a requirement for enrolment in this course. The number of participants is limited to n=20.

Literature

Articles form journals that published in the last couple of months (2-4 per student).
Purves Neuroscience (5th edition) as reference book,
Literature can be studied during the course and will be provided at the start of the course.

Behavioral Neurosciences

Course Code	AM_1299
Credits	6
Period	P2

Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M.C. van den Oever
Examiner	dr. P. Rao MSc
Teaching Staff	dr. M.C. van den Oever, prof. dr. R.E. van Kesteren, dr. N.J. Marchant, dr. D.R. Riga, dr. P. Rao MSc, dr. T. Pattij
Teaching method(s)	Seminar

Course Objective

Students will learn about widely-used animal models that allow a mechanistic dissection of behavioral phenotypes and neuronal circuits. Unique about this course is that we will have an in depth discussion of widely-used animal paradigms that model specific behavioral traits, as well as psychiatric and neurodegenerative disorders. The main aim is that students will acquire an in depth understanding of behavioral constructs and validity of these animal models. Related to this, students will learn about exciting new technological developments to study the neural underpinnings of behavior.

After the course, students will be able to:

1. Explain the behavioral read-outs of several well-established animal models of human behavior and brain disorders.
2. Determine and explain which behavioral paradigm and/or intervention technique should be used to address a specific research question.
3. Explain (and debate about) the advantages and limitations of the different models/techniques that have been discussed.
4. Write an animal protocol justified based on societal and scientific relevance and the three Rs (Replacement, Refinement, Reduction). An approved animal protocol is a mandatory first step to perform experimental animal research in The Netherlands and EU.

Course Content

Focus of the course is on widely used animal models that allow a mechanistic dissection of behavioral phenotypes and disorder symptoms at the level of neuronal circuits, cells and molecules. We will discuss paradigms of learning and memory, addictive behavior, impulsivity and depression, as well as genetic models (e.g. Alzheimer's disease), among others. We will demonstrate how these models can be integrated with state-of-the-art techniques, such as various viral vectors, optogenetics, chemogenetics and in vivo calcium imaging to dissect the neural correlates of behavior. In particular, advantages and limitations of the use of the different animal models and techniques will be explained and discussed.

In addition to lectures, students will debate with each other about the pros and cons of selected animal models and how they have contributed to a better understanding of normal and aberrant behavior.

Furthermore, we will teach students how to write a protocol for a license to perform animal experiments. In most countries, it is mandatory to motivate the scientific and societal relevance of the use of animals in research. Related to this, students will learn to provide a careful justification of the number of animals and the degree of discomfort that they will experience to address a specific research question.

Additional Information Teaching Methods

- Lectures (~20 h, mandatory)
- Debate sessions (~6 h, mandatory)
- Writing of protocol for animal experiments (mandatory).
- Tours through the mouse and rat facility at the VU (~3 h).

Students can use the remaining time for writing of the animal protocol and preparation of the debate sessions.

Method of Assessment

1) Exam with open questions about the lectures (40%)
2) Protocol for animal experiment (30%)
3) Debate sessions (30%)

For all three assessment forms, the minimal grade has to be 5.5 to pass the course.

Entry Requirements

Basic knowledge of animal behavior, physiology and neurobiology.

Literature

Mandatory exam material consists of the slides of all lectures and specific literature that will be provided on Canvas.

We will not make use of a textbook.

Additional Information Target Audience

2nd year VU Master Neurosciences students or 2nd year Master Biomedical Sciences student with track 'Neurobiology'.

2nd year students from a neuroscience related research master at a different university are eligible for enrollment, but should first contact the course coordinators, Dr. Michel van den Oever and/or Dr. Priyanka Rao.

Additional Information

There is a minimum of 5 participants in the course, and a maximum of 16 participants. Please email the course coordinator in case you have subscribed, but you are not going to attend the course.

Explanation Canvas

The final course schedule will become available on Canvas. Slides of the lectures will become available on Canvas. The written animal protocols have to be submitted to Canvas.

Recommended background knowledge

Basic knowledge of neurobiology

Career and Academic Skills and Portfolio year 2 Neurosciences

Course Code	AM_1268
Credits	3
Period	Ac. Year (September)
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	V. Chubar
Examiner	V. Chubar
Teaching Staff	V. Chubar, K. Wessels MSc, dr. J.I. Geertsema
Teaching method(s)	Study Group, Symposium

Course Objective

The Career & Academic Skills course intends to activate our students to think professionally and in an engaged manner about their study track, internships and future careers as neuroscientists.

Course Content

During year 2, the students will now put what they learned from the workshops and assignments from year 1 into practice by producing their own learning portfolio. Topics covered during year 2 themed workshops include: Pitching an idea or project successfully, job market orientation, job application & CV preparation, open science & data management. For the final portfolio, students need to justify, design, carry out, and reflect on their own self-directed professional development project (major project).

NB: This course runs across 2 academic years. All components from year 1 (AM_1250) must be completed & submitted in order to obtain 3 EC study credits during this year.

Additional Information Teaching Methods

Year 2 feedback workshops, self-guided professional development project (major project) work with mentors and/or

a tutor.

Method of Assessment

The 2nd year course consists of n=6 summative assignments that should be kept for inclusion in your final portfolio (summative) to demonstrate the 5 course learning components successfully.

Deadlines for course schedule will be communicated via the course manual & using CANVAS.

Entry Requirements

All students following the VU Master Neurosciences program are required to take this course.

Additional Information Target Audience

Second year VU Master Neurosciences students. This course is not open for external candidates.

Custom Course Registration

Workshops are given in small parallel groups of 10-15 students. Canvas instructions will follow at the beginning of the academic year. After the course has started, students should sign up for their workshop groups on Canvas, according to their own study timetable.

Explanation Canvas

Deadlines for the course schedule will be communicated via the course manual & using CANVAS.

Career and Academic Skills Portfolio year 1 Neurosciences

Course Code	AM_1250
Credits	0
Period	Ac. Year (September)
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	V. Chubar
Examiner	V. Chubar
Teaching Staff	dr. J.I. Geertsema, V. Chubar, K. Wessels MSc
Teaching method(s)	Seminar, Study Group, Symposium, Participatory Teaching Method, Lecture

Course Objective

The Career & Academic Skills course intends to activate our students to think professionally and in an engaged manner about their study track, internships and future careers as neuroscientists.

Course Content

During year 1, we aim to provide some context, preparation & feedback on the professional skills that scientists (in & outside of academia) are often required to use next to lab research; such as communication, peer-review, self-reflection, networking, project planning, and open science practice. We'll discuss these topics critically and practice them using in-class activities and small-scale assignments. Workshop topics include communication & feedback skills, peer-review, presentation skills, poster and scientific figure design, self-reflective practice, networking, goal-setting, and project planning. We'll discuss these topics, analyze examples, and practice both with in-class activities and small assignments.

NB: This course runs across 2 academic years. All components must be completed & submitted in order to obtain 3 EC study credits during year 2 (course code = AM_1268).

Additional Information Teaching Methods

Year 1 active workshops, Year 2 feedback sessions, self-guided project work with mentors.

Method of Assessment

This course consists of n=5 summative assignments that should be kept for inclusion in your final portfolio (summative) to demonstrate the 5 course learning components successfully.

Deadlines for course schedule will be communicated via the course manual & using CANVAS.

Entry Requirements

All students following the VU Master Neurosciences program are required to take this course.

Literature

Reading & preparation materials will be supplied in the CANVAS course at the beginning of first semester.

Additional Information Target Audience

First year VU Master Neurosciences students. This course is not open for external candidates

Custom Course Registration

This is a compulsory course for VU Master Neurosciences students.
Workshops are given in small parallel workgroups of 10-15 students. After the course has started, students can sign up for workshop groups on Canvas.
Canvas instructions will follow at the beginning of the academic year. After the course has started, students should sign up for workshop groups using Canvas, to a group that matches their own study schedule.

Cellular Models for Brain Disorders

Course Code	AM_1218
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. V.M. Heine
Examiner	prof. dr. V.M. Heine
Teaching Staff	prof. dr. V.M. Heine
Teaching method(s)	Study Group, Practical, Lecture

Course Objective

This course aims to introduce applied stem cell technologies. A rising number studies use induced pluripotent stem cell (iPSC) technologies to model human genetic brain disorders. The course illustrates experimental approaches to study brain disorders from genetic analysis, stem cell cultures to functional assays, such as genome-wide association studies (GWAS), iPSC cultures, neural cell differentiation, 3D organoid cultures, and functional iPSC product characterization. The use of these techniques is demonstrated by hands-on training and embedded in short research projects. Stem cell cultures will be analyzed and results will be presented and discussed in plenary sessions. Current state of genetic and stem cell field in neuroscience will be explained in supporting lectures.

Learning objectives:
1. The student understands how human genetic studies can guide iPSC research.
2. The student understands the prospects and challenges of stem cell technologies.
3. The student is familiar with current state of iPSC research in the context of genetic brain diseases, such as neuropsychiatric disorders.
4. The student can follow procedures for good cell culture practice.
5. The student can define a specific research question in a small team and translate this into an experimental set-up with appropriate controls.
6. The student can process obtained experimental data in a clear oral presentation in English. The student contributes actively to the discussion of the other presented projects.

Course Content

This is a practical course. In small work groups you will be introduced to advanced stem cell technologies and receive hands-on training. By executing a small research project, multi-disciplinary research involving genetic studies and stem cell assays, will be illustrated. Different stem cell cultures, including iPSC and 3D organoid cultures, will be demonstrated. Theoretical background and methods will be achieved by using examples from

recent literature.

Additional Information Teaching Methods

lecture, practicals, work groups and presentations.

Participation in all practicals/workgroups is obligatory. Attending the lectures is highly recommended.

Method of Assessment

The grading will be as follows:
Final exam (lecture content): 30%
Research Report: 20%
Research Presentation: 20%
Practicals / Lab Execution: 30%

To pass the course one needs a grade that is at least 5.5 for every part; compensation is not possible.

Entry Requirements

Are you enrolled in another program than MSc Neurosciences and would like to participate in this course? Send an e-mail to the course coordinator before the course starts so that your eligibility can be evaluated.

Literature

Literature can be studied during the course and will be provided at the start of the course.

Additional Information Target Audience

2nd year VU Master Neurosciences students. Are you enrolled in another program and would like to participate in this course? Send an e-mail to the course coordinator before the course starts so that your eligibility can be evaluated.

Additional Information

A maximum of 15 students can be accommodated in the course. Preference is given to students who participate in the VU Master Neuroscience program. Course is taught in English.

Clinical neurosciences

Course Code	AM_1005
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	V. Donega
Examiner	V. Donega
Teaching Staff	dr. C. Vriend, V. Donega, dr. D.P. Bakker
Teaching method(s)	Practical, Written partial exam, Lecture

Course Objective

The aim of this course is twofold:

1. Provide knowledge on the clinical features, pathophysiology, molecular and cellular mechanisms, and latest developments in therapeutic strategies for a select number of neurological and neuropsychiatric disorders.
2. Initiate a translational way of thinking as a neuroscientist.

This course will provide the latest clinical and neurobiological knowledge on a number of key brain disorders: Multiple Sclerosis, Childhood White Matter Disorders, Parkinson’s disease, Neuropsychiatric Disorders, and Glioma.

It also introduces some of the tools that researchers have at their disposal to investigate the pathophysiology of brain disorders in vivo, such as neuroimaging.

This course will stimulate translational thinking by continuously crosslinking clinical features, (histo)pathology, and imaging abnormalities for diagnosis, prognosis, and therapeutic strategies.

Given the diversity of subjects and to provide the most up-to-date insight into the pathophysiology of the brain disorders, lectures are provided by both (external) pre-clinical and clinical researchers that are at the top of their respective fields.

Course Content

The course evolves around the following brain disorders and diseases Multiple Sclerosis, Childhood White Matter Disorders, Parkinson's disease, Neuropsychiatric Disorders, and Glioma.

Multiple sclerosis:
Description of the clinical phenotypes of the disease. Introduction of neuroimaging specifically adapted for MS. The use of neuropathology to understand disease mechanisms in MS.

Childhood white matter disorders (CWMD):
Introduction to the main clinical manifestations and different forms of CWMD. It will provide an example of translational research: from patient to gene, protein, and treatment.

Movement disorders:
The focus will be on Parkinson's disease. Description of the motor and cognitive manifestations of movements Disorders, and their underlying neurobiology. The latest research on the underlying molecular mechanisms, and therapeutic strategies will be discussed.

Neuropsychiatric disorders:
Description of the clinical manifestation of some of the main neuropsychiatric disorders, such as obsessive-compulsive disorder, major depressive disorder, and post-traumatic stress disorder. Introduction to the underlying neuroanatomy and pathophysiology and the role of neuroimaging in neuropsychiatry. The latest therapeutic treatments will be discussed.

Neuro-oncology:
Description of the clinical/cognitive manifestations of brain tumors. The focus will be on glioma. The effects of these tumors on brain networks is the main focus.

The course will additionally assess the students' skills to:
Summarize and present the essence of a scientific paper in the form of a graphical abstract and an oral presentation. Students are expected to participate actively during these assignments.

Additional Information Teaching Methods

Lectures, student presentations, and group assignments (live on campus). The course runs for eight weeks, two days per week.

Method of Assessment

- Group assignment to produce a graphical abstract on a recent cutting-edge paper in the field of Clinical Neuroscience (10% of grade);
- Group presentation on recent cutting-edge paper in the field of Clinical Neuroscience (30% of grade);
- A written digital exam with (primarily) open questions about all the course material (60% grade).

Sufficient grades (>5.5) for the presentation, as well as the exam are required in order to pass the course. Individual grades cannot be averaged to obtain a sufficient grade.

Entry Requirements

All students following the Neurosciences masters program at the VU are required to take this course.

This course is not open for external candidates.

Literature

Course slides that will be provided (exam material) and research papers (assignment material).

Additional Information Target Audience

MSc Neuroscience students;

Additional Information

Questions? clinicalneuroscience@amsterdamumc.nl

This course is coordinated by: dr. V. Donega (Vanessa) and dr. D.P. Bakker (Dewi)

Custom Course Registration

Enrolment through vu-net.

Recommended background knowledge

Knowledge on neuroanatomy, (basic knowledge) brain disorders, and basic knowledge on cell biology is**highly** recommended. Recommended literature: Neuroanatomy: an illustrated colour text (5th edition, Crossman & Neary, ISBN 9780702054051) and Molecular Biology of the Cell (Alberts et al., ISBN-10: 0-8153-4072-9).

Developmental Neurobiology of the Vertebrate Brain

Course Code	AM_470713
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. R.F.G. Toonen
Examiner	prof. dr. R.F.G. Toonen
Teaching Staff	prof. dr. R.F.G. Toonen
Teaching method(s)	Lecture

Course Objective

Many brain disorders stem from disruptions in early brain development. To effectively tackle these conditions, it is essential for future neuroscience researchers to acquire a deep and comprehensive understanding of how the brain develops. This course aims to provide students with detailed insights into the cellular and molecular mechanisms that drive neural circuit formation and plasticity during brain development. Key topics include:

- Molecular pathways underlying neurogenesis, neuronal migration, differentiation, axonal growth, and synapse formation
- The concept of critical periods in development and mechanisms of experience-dependent plasticity
- The relationship between developmental processes and the emergence of brain disorders
- Brain plasticity during both embryonic and adult stages

Throughout the course, these subjects will be discussed in the context of state-of-the-art research techniques and experimental approaches, preparing students to critically assess and contribute to the field of developmental neuroscience.

Course Content

The course is structured into three main components: lectures, master classes, and hands-on training.

During the first two weeks, students will attend in-depth lectures covering key topics in developmental neurobiology, including neurogenesis, neuronal migration, differentiation, axonal growth, synaptogenesis, and critical periods of plasticity. These will be complemented by master classes led by experts in the field, offering deeper insights into current research and emerging concepts in brain development.

The third and fourth weeks will focus on practical, hands-on training involving early neuronal development in both mouse and human model systems. Students will gain experience with experimental techniques commonly used in developmental neuroscience research.

In the final week, students will give presentations on selected topics in developmental neurobiology, demonstrating their understanding of the material and their ability to critically evaluate current literature.

Additional Information Teaching Methods

This is a full time course.
Week 1&2: Lectures and seminars. These 2 weeks will be examined during a mid-term exam.
Week 3&4: Master classes, Journal Clubs and hands-on practicals focused on early development of neuronal

networks, Human iPS cells and neuro- glia interactions. In week 4, students will present on specific topics in developmental neurobiology.

Theory (30%) A. Lectures - 8 sessions of ± 3 hrs: 24 hrs B. Masterclasses - 3 classes of ± 3 hrs: 9 hrs C. Individual Journal Club task - 1 high-impact paper presentation 6 sessions of 2 hrs: 12 hrs Hands-on analysis (30%) of neuron development in vitro (polarization, neurite outgrowth, synapse formation): 10 sessions of ± 4 hrs: 40 hrs - Presentation of results lab work: 1 session of 5 hrs Self study (40%) - 60 hrs

Method of Assessment

Written mid-term exam (40%); Oral presentation of seminar task (30%); Presentation of labwork (30%). Students need to pass all parts (grade > 5.5) to obtain final grade.

Literature

Handouts will be distributed at the beginning of the course. PDFs of all lectures will be made available via Canvas.

Additional Information Target Audience

2nd year VU Master Neurosciences students.

Custom Course Registration

The course has a maximum capacity of 25 students. Priority will be given to students enrolled in the VU Master Neuroscience program. Students from other Master's programs may be admitted based on availability. All students who are not part of the Master of Neuroscience program **must** contact the course coordinator, Prof. Ruud Toonen (r.f.g.toonen@vu.nl), prior to enrollment. Please include details of your current study program to determine your eligibility and course availability.

Experimental and Clinical Neuroendocrinology

Course Code	AM_470700
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Stenvers
Examiner	dr. D.J. Stenvers
Teaching Staff	dr. D.J. Stenvers
Teaching method(s)	Practical, Lecture

Course Objective

At the end of this course the student can demonstrate an understanding of lectured experimental and clinical neuroendocrinology and describe recent developments and current research in the field of clinical neuroendocrinology.

Course Content

Clinical neuroendocrinology focuses on the pathophysiology, diagnosis, and treatment of diseases of the hypothalamus and pituitary gland. The course includes lectures on the structure and function of the hypothalamus and pituitary gland, with special emphasis on central regulation of growth, puberty, reproduction, metabolism, obesity and stress, sexual orientation and gender identity, taking both an experimental and clinical point of view.

Additional Information Teaching Methods

Lectures 24 hrs (appr)
Research tutorials 5-10 hrs (appr)

Method of Assessment

50% Examination with multiple choice and open questions derived from the lectures, provided literature and research tutorials.

50% Assignment that consists of a(n) (audio-visual) presentation (knowledge clip/vlog/graphic novel/e-learning) explaining a specific subject related to the lectures.

The final result will be the average of the examination- and the assignment grade.

Please note that both the examination and the assignment result must be sufficient (at least 5.5).

Literature

Selected literature and scientific papers will be provided during the course.

Additional Information Target Audience

2nd year VU Master Neurosciences students.

Additional Information

Maximum number of participants is 30. For more information, please contact drs. K.A.J. Smit (k.a.j.smit@amsterdamumc.nl).

Final Year Project Neurosciences

Course Code	AM_1269
Credits	30
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M.D. Zwan
Examiner	dr. C. de Boer
Teaching Staff	
Teaching method(s)	Other

Course Objective

The aim of a Final Year Project is to learn how to independently conduct neuroscientific research in a 6-month internship.

Course Content

After completing your internship, you will be able to:

1. Connect to researchers in the field and prepare and discuss an original research plan.
2. Find scientific information independently and evaluate this to formulate a clear hypothesis.
3. Independently design an experiment and/or collect relevant data to address your research question. This includes, but is not limited to, the use of relevant compute software, command relevant laboratory procedures, use appropriate safety measures, and command social-and practicalskills to collect data in human subjects.
4. Critically analyse and interpret the data collected as part of your research project.
5. Write a discussion according to academic standards. This includes an critical interpretation of your data, speculate on the implications of your findings, and reflection on the ethical aspects of your research, all within the context of current literature of your topic.
6. Communicate your experimental results in an oral presentation and engage in scientific discussions on your findings.
7. Display a good/mature level of general academic competencies, including planning and time management, having a scientifically critical attitude, verbal discussion skills, and your ability to self-reflect on your performance.

Additional Information Teaching Methods

The supervisor teaches the student the necessary neuroscientific research techniques/tools and analysis tools and regularly provides feedback on the performance of the student. The VU supervisor or on-site supervisor also aids in the student's development of the following skills: academic skills (incl. insight and creative thinking), scientific writing skills, presenting skills, attitude.

Method of Assessment

Formative assessment - Written research proposal prior to the start of the internship, Go/No-Go evaluation ~6 weeks after the start of the internship (decision on whether the project and the student both have enough potential to continue).

Summative assessment - Final grade: attitude & execution (40%), oral presentation (20%), written report (assessor 1: 20%; assessor 2: 20%).

If the student fails to upload any report before the agreed report deadline, the maximum partial grade for the report cannot be higher than 6,5.

A third assessor will be appointed by the Neurosciences Examination Board to assess the final report in the case that:

- the difference between the grade of the final report by the VU supervisor and the second assessor is equal to or more than 1,5 (on a scale 1-10). The final grade will then be calculated as an average from all three.
- One of the assessors evaluates the report as insufficient. The third assessor will determine whether the report is sufficient or not. If sufficient, the final grade will be the average of the highest two grades. If insufficient, the final judgment of the report will be insufficient and the student has failed the assessment.

Entry Requirements

You need to have successfully completed Internship Neurosciences I and obtained at least 18EC of coursework prior to start of the Final Year Project.

Literature

Starting literature is depending on the topic of the project and therefore provided by the supervisor.

Additional Information Target Audience

Year 2 Master Neurosciences students

Additional Information

The Final Year Project is considered the end product of the programme. Therefore, no exemptions will be granted (See Teaching and Examination Regulations Part B).

Custom Course Registration

Course enrolment via VU-net.

NB: the project needs prior approval of the Internship coordinator. The student uploads the Research proposal form via Onstage, preferably at least two weeks prior to the start date of the internship.

From Molecule to Mind

Course Code	AM_1275
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	D.A. Maas
Examiner	prof. dr. H.D. Mansvelder
Teaching Staff	dr. G.J. Schenk, prof. dr. M. Verhage, prof. dr. H.D. Mansvelder, D.A. Maas, prof. dr. A.B. Smit
Teaching method(s)	Practical, Written partial exam, Lecture

Course Objective

Providing master students, independent of primary training, with a solid basis in molecular and cell biology, neurophysiology and functional neuroanatomy.

Course Content

The course will be used to brush up your knowledge of neurophysiology, molecular and cellular neurobiology and neuroanatomy. Lectures by established researchers on exciting and cutting-edge topics as well as a keynote symposium related to the different research programs on campus will be held to give you an impression of the research we do and to prepare you for your search for an internship project (which will start in February 2024).

Additional Information Teaching Methods

Lectures, study groups, assignments, practical sessions*, presentations, peer feedback, demonstrations.

*Please be aware the number of spots for the practicals is limited. In case the maximum number of students is exceeded, the course coordinator will decide who will be registered for the practicals, for the others there is an online practical available.

Method of Assessment

You will have to prepare presentations based on neuroanatomy related assignments and are expected to participate actively during practical sessions. You will receive a pass/fail grade for neuroanatomy. A 'pass' will be rewarded by a 30% correct starting grade for the final exam; an 'incomplete' will be rewarded by 0% (which means your maximum final grade for the course will be 70%). The final exam will consist of a digital multiple choice examination with two parts: cell biology and neurophysiology.

Entry Requirements

All students following the VU Master Neurosciences program are required to take this course.

Literature

- Alberts et al., Essential Cell Biology, 4th edition (2013; ISBN 9780815344544). Purves et al., Neuroscience, 5th edition (2012; ISBN 9780878936953). NB. More recent print editions of these books may also be used.

Additional Information Target Audience

First year VU Master Neurosciences

Recommended background knowledge

We strongly advise you to study these chapters before the course starts.NB. Alternatively, study equivalent chapters from more recent printings of these books:

- Alberts et al., Essential Cell Biology, 4th edition (2013; ISBN 9780815344544). Chapters 4, 5, 7, 8, 15 - 17
- Purves et al., Neuroscience, 5th edition (2012; ISBN 9780878936953). Chapters: 3 - 8.

Functional Brain Imaging

Course Code	AM_470715
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. P.J.W. Pouwels
Examiner	dr. P.J.W. Pouwels
Teaching Staff	
Teaching method(s)	Computer lab, Lecture

Course Objective

The student will learn about the most important brain imaging techniques, the principles on which they are based and practical applications in research and patient care. Quite some emphasis on physics and mathematics will be given during the lectures. However, the goal is not to examine the students on this knowledge, but to provide sufficient background to develop a critical and scientific attitude towards imaging techniques. The main focus for the students will be to translate this knowledge into the potential use of techniques in neuroscience.

Course Content

In this course the student will learn about techniques to study structure and function of the brain that are used in clinical studies and in neuroscientific research. Three main approaches can be distinguished: neurophysiological techniques (EEG, MEG), magnetic resonance techniques (MRI, fMRI, MRS) and techniques which involve the use of radio active ligands (SPECT, PET). In addition, network approaches to describe brain structure and function will be discussed. The background, possibilities and limitations of these techniques will be discussed in relation to ongoing research at the Amsterdam UMC and Amsterdam Neuroscience, including Brain imaging (methodological innovation), Brain mechanisms (childhood white matter disorders), Neurodegeneration (Alzheimer's disease, Parkinson's disease), Neuroinfection & neuroinflammation (Multiple Sclerosis), Compulsivity, impulsivity & attention (obsessive compulsive disorder).

Additional Information Teaching Methods

The basic principles and several applications of all techniques will be presented in a series of lectures scheduled during the first 6 weeks (typically two or three days a week - the other days are available for Advanced Clinical Neuroscience). Attendance at the lectures is required. Small groups will discuss particular aspects in more detail. During the course, visits to the departments involved in imaging will be arranged. Hands-on experience of analysis methods is provided in computer practicals.

Method of Assessment

Individual exam (50% of final mark).

Team presentation about a functional brain imaging experiment about a neurological disorder / neuroscientific problem. (50% of final mark).

Both components have to be awarded a sufficient grade in order to receive a sufficient final grade.

Literature

Recommended background literature:
- Bullmore E, Sporns O: Complex brain networks: graph theoretical analysis of structural and functional systems. Nat Rev Neurosci. 2009 Mar;10(3):186-98.
- Gong G et al: Brain connectivity: gender makes a difference. Neuroscientist. 2011 Oct;17(5):575-91
- Stam CJ: Characterization of anatomical and functional connectivity in the brain: a complex networks perspective. Int J Psychophysiol. 2010 Sep;77(3):186-94.
- Stam CJ: Modern network science of neurological disorders. Nat Rev Neurosci. 2014 Oct;15(10):683-95.
- Varela F et al: The brainweb: phase synchronization and large-scale integration. Nat Rev Neurosci. 2001 Apr;2(4):229 39.
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- Vidaurre D et al: Brain network dynamics are hierarchically organized in time. Proc Natl Acad Sci U S A. 2017 Nov 28;114(48):12827-12832.
- Devlin H et al: Introduction to fMRI. http://www.fmrib.ox.ac.uk/education/fmri/fmri/introduction-to-fmri
- Matthews PM et al: Applications of fMRI in translational medicine and clinical practice. Nat Rev Neurosci. 2006 Sep;7(9):732-44.
- Lazar M: Mapping brain anatomical connectivity using white matter tractography. NMR Biomed. 2010 Aug;23(7):821-35.
- Beaulieu C. The basis of anisotropic water diffusion in the nervous system - a technical review. NMR Biomed. 2002 Nov-Dec;15(7-8):435-55
- Blokland JA et al: Positron emission tomography: a technical introduction for clinicians. Eur J Radiol. 2002 Oct;44(1):70-5.
- Klunk WE et al: Imaging brain amyloid in Alzheimer's disease with Pittsburgh Compound-B. Ann Neurol. 2004 Mar;55(3):306-19.
- Lammertsma AA: Radioligand studies: imaging and quantitative analysis. Eur Neuropsychopharmacol. 2002 Dec;12(6):513-6.
- Phelps ME: Positron emission tomography provides molecular imaging of biological processes. Proc Natl Acad Sci U S A. 2000 Aug 1;97 (16):9226-33.
- Villemagne VL et al: Aβ Imaging: feasible, pertinent, and vital to progress in Alzheimer's disease. Eur J Nucl Med Mol Imaging. 2012 Feb;39(2):209-19.
- Villemagne VL et al: Tau imaging: early progress and future directions. Lancet Neurol. 2015 Jan;14(1):114-24.
- Finnema SJ et al: Imaging synaptic density in the living human brain. Sci Transl Med. 2016 Jul 20;8(348):348ra96.

-Kreisl WC et al: PET imaging of neuroinflammation in neurological disorders. Lancet Neurol 2020 Nov;19(11):940-50.

- Meyer JH et al: Neuroinflammation in psychiatric disorders: PET imaging and promising new targets. Lancet Psych 2020 Dec; 7(12): 1064-74.
- Baillet S et al: Electromagnetic brain mapping. IEEE Signal Processing Magazine 2001 Nov;18(6):14-30.

- Hari R, Salmelin R: Human cortical oscillations: a neuromagnetic view through the skull. Trends Neurosci. 1997 Jan;20(1):44-9.
- Hillebrand A et al: A new approach to neuroimaging with magnetoencephalography. Hum Brain Mapp. 2005 Jun;25(2):199-211.
- Boto E et al: Moving magnetoencephalography towards real-world applications with a wearable system. Nature. 2018 Mar 29;555(7698):657-661.

Additional Information Target Audience

2nd year VU Master Neurosciences students.

Additional Information

The maximum number of participants is 30. This means that students other than the target group should first

contact the coordinator. Taught in English.

For further information, please contact dr. P.J.W. Pouwels - (Petra) - (pjw.pouwels@amsterdamumc.nl)

Custom Course Registration

Students not enrolled in the VU Master of Neuroscience are required to contact the course coordinator prior to enrollment. The course coordinator determines whether you can participate.

Genetics in Neuroscience

Course Code	AM_1214
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	J.E. Savage
Examiner	J.E. Savage
Teaching Staff	J.E. Savage
Teaching method(s)	Computer lab, Written partial exam, Lecture

Course Objective

After completing this course, students should be able to:

- Explain theoretical principles underlying human genetics
- Describe current state of the art methods for gene discovery for brain-related traits
- Apply computational methods to analyze large genetic datasets
- Compare tools that can be used to gain biological insight into genetic findings
- Critically evaluate results from genetic discovery studies of complex traits
- Reflect on how genetics and neuroscience can inform each other
- Apply genetic methods to inform the design of neurobiological studies

Course Content

The course is offered to students in the Master of Neurosciences. The main goal is to teach how statistical genetic studies are conducted to identify genes for complex neuropsychiatric traits and disorders. The course will discuss how results of such studies should be interpreted and how they can be used to inform neurobiological research. The lectures will provide a comprehensive overview of the current revolution in the field of human genetics, and will also lay out the theoretical foundations of this field. In addition, the lectures aim to provide starting points for bridging human genetics and neurobiology. Lectures will include examples of how to apply statistical and bioinformatic tools to gain insight into the genetics of brain-related traits. Practical sessions are aimed at providing hands-on experience with current analytical tools and fostering a critical opinion on published studies in this field.

Additional Information Teaching Methods

13 lectures (2 hours each), 10 computer practical sessions (3 hours each), and 1 presentation session.

Contact hours: 57 hours distributed over 24 sessions. Students are expected to read scientific papers to prepare for

each lecture.

Lecture material: Lecture slides + scientific papers. Active participation in lectures and discussion sessions is expected.

Computer practical sessions consist of: - Unix programming - practical exercises using genetic tools (e.g. PLINK, FUMA, MAGMA, PRSice), covering: - quality control of genetic data - running a genome-wide association study - interpreting the results from a genome-wide association study in biological context - conducting genetic risk prediction based on the results from a genome-wide association study. Computer assignments need to be completed during the computer practical session or handed in via Canvas. Students will gain hands-on experience with the analysis pipeline used in genome-wide association studies.

Assignments and presentations will involve critical evaluation of scientific literature and application of statistical genetic methods to neuroscience research.

Literature: Scientific papers that will be provided throughout the course.

Method of Assessment

One final exam in week 8, one individual essay (graded) in week 4, one group presentation (graded) in week 7, and assignments for each computer practical. The final score is the weighted average of .65*exam + .10*essay + .20*presentation + .05*practicals. A sufficient grade (5.5. or higher) is required for both the exam and the overall score in order to pass the course.

Practical attendance is a necessary condition to pass this course. Practical sessions and presentations are mandatory.

Resit: Offered for the exam.

Entry Requirements

All students following the VU Master Neurosciences program are required to take this course.

Literature

Scientific articles that will be provided throughout the course.

Additional Information Target Audience

First year VU Master Neurosciences students. This course is not open for external candidates.

Recommended background knowledge

Basic knowledge of genetics and/or cell biology, neuroscience or psychology, and familiarity with basic computer programming is recommended.

Internship Neurosciences I

Course Code	AM_471108
Credits	27
Period	Ac. Year (September)
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M.D. Zwan
Examiner	dr. C. de Boer
Teaching Staff	
Teaching method(s)	Other

Course Objective

The aim of Internship I is to learn to conduct neuroscientific research in a 5-month internship.

Course Content

After completing your internship, you will be able to:

1. Connect to researchers in the field and together formulate and discuss an original research plan.
2. Find scientific information and evaluate this to formulate a clear hypothesis.
3. Design an experiment and/or collect relevant data to address your research question. This includes, but is not limited to, the use of relevant compute software, command relevant laboratory procedures, use appropriate safety measures, and command social-and practical skills to collect data in human subjects.
4. Analyse and interpret the data collected as part of your research project.
5. Write a discussion according to academic standards. This includes an interpretation of your data, speculate on the implications of your findings, and a reflection on the ethical aspects of your research.
6. Communicate your experimental results in an oral presentation and engage in scientific discussions on your findings.
7. Display a sufficient level of general academic competencies, including planning and time management, having a scientifically critical attitude, verbal discussion skills, and your ability to self-reflect on your performance.

Additional Information Teaching Methods

The supervisor teaches the student the necessary neuroscientific research techniques/tools and analysis tools and regularly provides feedback on the performance of the student. The VU supervisor or on-site supervisor also aids in the student's development of the following skills: academic skills (incl. insight and creative thinking), scientific writing skills, presenting skills, attitude.

Method of Assessment

Formative assessment - Written research proposal prior to the start of the internship, Go/No-Go evaluation ~6 weeks after the start of the internship (decision on whether the project and the student both have enough potential to continue).

Summative assessment - Final grade: attitude & execution (40%), oral presentation (20%), written report (assessor 1: 20%; assessor 2: 20%).

If the student fails to upload any report before the agreed report deadline, the maximum partial grade for the report cannot be higher than 6,5.

A third assessor will be appointed by the Neurosciences Examination Board to assess the final report in the case that:

- the difference between the grade of the final report by the VU supervisor and the second assessor is equal to or more than 1,5 (on a scale 1-10). The final grade will then be calculated as an average from all three.
- One of the assessors evaluates the report as insufficient. The third assessor will determine whether the report is sufficient or not. If sufficient, the final grade will be the average of the highest two grades. If insufficient, the final judgment of the report will be insufficient and the student has failed the assessment.

Entry Requirements

You need to have obtained at least 12 ECTS from the MSc Neurosciences first year programme prior to start of the internship.

Literature

Starting literature is depending on the topic of the project and therefore provided by the supervisor.

Additional Information Target Audience

Year 1 MSc Neurosciences students.

Additional Information

Note that the (first part of the) internship runs in parallel with the course 'Academic Writing: Neuroscience (AM_1123)', for which (weekly) mandatory in-person attendance is required (more detailed information can be found at the 'Academic Writing: Neuroscience' Canvas page). Therefore, responsibilities as a student are to enroll in the Canvas course as soon as possible, notify your Internship supervisors of this mandatory course and associated (attendance) requirements, and integrate writing sessions into your weekly project planning.

Custom Course Registration

Course enrolment via VU-net.

NB: the project needs prior approval of the Internship coordinator. The student uploads the Research proposal form via Onstage, preferably at least two weeks prior to the start date of the internship.

Live Cell Imaging

Course Code	AM_470726
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. R.F.G. Toonen
Examiner	prof. dr. R.F.G. Toonen
Teaching Staff	dr. M. Koppers, prof. dr. R.F.G. Toonen
Teaching method(s)	Lecture

Course Objective

This course will provide the student with theoretical and practical knowledge to utilize emerging cellular and sub-cellular imaging technologies in neuroscience.

Course Content

The recent advances in light microscopy, digital image processing, and the development of innovative fluorescent probes have greatly expanded the possibilities for studying the nervous system—ranging from individual synaptic terminals to complex brain networks. This intensive theoretical and practical course offers students in-depth knowledge of cutting-edge imaging technologies, with a primary focus on live-cell (vital) light microscopy. Participants will gain a solid understanding of the principles of light microscopy and hands-on experience with a variety of tools and techniques, including:

- Advanced light microscopy systems
- Electronic cameras and laser scanning technologies
- Functional fluorophores and their delivery methods
- Digital image acquisition and analysis software

The course will also cover transmitted light microscopy for visualizing cellular dynamics, as well as molecular probes that report on cell function. These include calcium and voltage sensors, photoactivatable (“caged”) compounds, and markers of exocytosis and vesicle trafficking. Finally, students will explore the challenges and opportunities involved in integrating imaging techniques with electrophysiological approaches, preparing them for multidisciplinary research in modern neuroscience.

Additional Information Teaching Methods

This is a full time course. In the first two weeks we will address all major live cell imaging techniques and their applications in a series of lectures and Masterclass meetings with experts in the field. A mid-term exam will test the obtained knowledge. The last 2 weeks will be devoted to hands-on experiments in the lab in small groups. Students will perform 2 imaging experiments under guidance of an experienced PhD- student or Postdoc.

Method of Assessment

Mid-term Exam (50%)

Two oral group presentations at the end of each experimental week (50%)

Students need to pass all parts (grade > 5.5) to obtain final grade.

Literature

Course coordinators will provide selected chapters from Live Cell Imaging. A laboratory Manual. Editors: Goldman and Spector and a selection of primary scientific papers at start of the course.

Additional Information Target Audience

2nd year VU Master Neurosciences students.

Custom Course Registration

Course is also open to 2nd year Master students from other courses and to non-VU neurosciences students but **ONLY** if course is not filled-up by VU Master students (max. capacity is 25 students). Non-Neurosciences master students **NEED** to contact course coordinator (Ruud Toonen, r.f.g.toonen@vu.nl) with study program details for eligibility check **prior** to self enrolment.

Network Neuroscience

Course Code	AM_1316
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. M.P. van den Heuvel
Examiner	prof. dr. M.P. van den Heuvel
Teaching Staff	prof. dr. M.P. van den Heuvel, dr. R.M. Brouwer
Teaching method(s)	Study Group, Lecture

Course Objective

Network Neuroscience is an interdisciplinary field that applies network science to study the brain's structure and function. It models the brain as a complex network of interconnected regions, using graph theory and computational methods to analyze neural connections at different scales—from individual neurons to large-scale brain regions. This approach helps researchers understand cognitive functions, brain disorders, and how neural circuits process information. Network neuroscience has applications in brain mapping, mental health research, and artificial intelligence, offering insights into how the brain organizes and adapts over time. By examining how brain networks are organized and how they change over time, network neuroscience contributes to advancements in brain mapping, mental health research, and artificial intelligence.

In this course students will interactively learn and experience the exciting new field of Network Neuroscience. Students will:

- Understand Brain Network Reconstruction – Learn the fundamental techniques for reconstructing brain networks from health and clinical MRI data.
- Master Graph Theory and Network Science – Develop a strong foundation in the mathematics, applied statistics, and computational models used in Network Neuroscience.
- Apply Network Neuroscience Techniques – Gain hands-on experience in analyzing real-world MRI data using advanced network models and statistical methods.
- Develop Programming and Computational Skills – Learn to implement and apply network analysis methods using programming tools relevant to neuroscience (coding : MATLAB)
- Conduct Independent Data-Driven Research – Design and execute a personal research project using real data, leading to novel scientific insights.
- Enhance Critical Thinking and Scientific Communication – Critically evaluate computational neuroscience techniques and communicate findings effectively through a scientific paper.

Course Content

The course will include interactive lecture sessions, computer lab sessions, and experimental time working on the final project.

Lectures will include 20% of the study time of the course.

Practicals will include practical sessions supervised by the teacher, and practical sessions involving peer-review and self-study working on their the data project, together around 30 and 50% of the study time of the course.

Additional Information Teaching Methods

All course material will be presented on a dedicated CANVAS page

Method of Assessment

- A written final exam (Multiple Choice and/or open questions regarding network neuroscience fundamentals and programming)

- Weekly assignments : pass/fail; normative test to monitor progress. Assignments will be discussed in flipping the class-room sessions
- Final report

Additional Information Target Audience

2nd year VU Master Neurosciences students.

Explanation Canvas

All course material will be presented on a dedicated CANVAS page

Neuro- and Psychopharmacology

Course Code	AM_470718
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M.M.M. Wilhelmus
Examiner	dr. M.M.M. Wilhelmus
Teaching Staff	dr. T. Pattij, dr. B. Drukarch, dr. M.M.M. Wilhelmus, dr. G.J. Schenk
Teaching method(s)	Written partial exam, Lecture

Course Objective

Principal aim of the course is to provide the student with sufficient insight into the basic principles of Neuro- and Psychopharmacology, in order to enable the student to independently formulate ideas and concepts which may lead to the development of innovative drug therapies. The subjects of the course are largely drawn from research themes of Neuroscience Amsterdam.

Learning objectives:

- Students are able to describe the pathophysiology of the disease under study and know which biological mechanisms are the driving forces underlying the pathological process(es).
- Students are capable of pinpointing possible therapeutic targets in the disease process(es), and can argue whether and how the therapy will modulate these therapeutic targets.
- Students are able to defend and motivate their therapeutic strategy in front of fellow students and teachers.

Course Content

Current pharmacotherapy of well known neurologic- and psychiatric disorders is based on the use of drugs the development of which has been guided by, often unproven, hypotheses concerning the pathophysiology of these diseases. As a result, despite a few "success stories", in most cases this has led to the introduction of suboptimally effective compounds. Moreover, these "therapeutics" do not cure the disease, but merely act to ameliorate the symptoms. However, in light of the increasing knowledge concerning the pathogenesis and pathophysiology of brain disorders, it now seems highly feasible that in the near future drugs will be developed which will target the core(s) of the disease process. During the course, a small number of lectures will provide the student with knowledge and insight into the hypotheses that have guided the development of a large number of the current generation of neuro- and psychopharmaceuticals. Subsequently, the student will be expected to 1) independently identify the strong points and weaknesses of these hypotheses, and 2) use the conclusions of this evaluation, in combination with recent developments in the field, to formulate a well-based proposal for the development of a new pharmacotherapy for a specific brain disease.

Additional Information Teaching Methods

Lectures (approximately 12h), 29 knowledge clips (self-study), Question & Answer sessions (4h), progress meeting with supervisor (approximately 3h), pitch your idea (2h Pitch session), and writing a thesis and final presentation of the thesis (approximately 96h). In total, the students are expected to invest a workload of 20 hours each week during this course.

Method of Assessment

Pharmacology assignment (15% of final grade), thesis (50% of final grade) and public presentation (35% of mark)

of thesis.

Literature

- Drukarch & Wilhelmus's Essentials of Pharmacology. B. Drukarch and Micha M.M. Wilhelmus. Second edition, 2024, Amsterdam, The Netherlands. ISBN number: 978-90-9030859-3.
- Additional texts provided during the course

Additional Information Target Audience

2nd year VU Master Neurosciences students.

Additional Information

A maximum of 24 students can be accommodated in the course. Preference is given to students who participate in the VU Master Neurosciences program. Course is taught in English. For all 3 assessment elements (pharmacology assignment, the thesis and final presetation), no resit or alternative assignments are in place. For further information please contact dr. M. Wilhelmus (m.wilhelmus@amsterdamumc.nl).

Recommended background knowledge

(Medical) Pharmacology course or equivalent at bachelor level.

Neurogenomics

Course Code	AM_1007
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	B.M.W. Borgman
Examiner	prof. dr. A.B. Smit
Teaching Staff	prof. dr. A.B. Smit, prof. dr. M. Verhage
Teaching method(s)	Computer lab, Study Group, Lecture

Course Objective

To provide the Master of Neuroscience students with a solid basis in understanding the genome and working mechanisms and function of genes in relation to the development and functioning of the nervous system.

Course Content

The course will address the various aspects of functional analysis of the genome, by addressing the following topics:

- The search for genes and gene variants which are underlying neuronal physiology and pathology, including forward genetics and gene-hunting strategies
- Functional analysis of genes through reverse genetics
- Gene expression analysis of neuronal cells and brain areas (gene expression profiling)
- The analysis of proteins (proteomics) and complexes thereof.
- Simulation of genetic and protein networks
- The introduction of various model organisms relevant for neurogenomics research, such as, man, mouse, Drosophilla, C. elegans, and zebrafish.

Additional Information Teaching Methods

Lectures, masterclasses, workshops, student presentations

Method of Assessment

Presentation: 20% and written examination: 80%.

Entry Requirements

All students following the VU Master Neurosciences program are required to take this course.

Literature

To be announced on Canvas

Additional Information Target Audience

First year VU Master Neurosciences students. This course is not open for external candidates.

Additional Information

Language: tuition in English

Custom Course Registration

Students need to enroll via VUnet.

Neuron-Glia Interactions

Course Code	AM_1217
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M.H.G. Verheijen
Examiner	dr. M.H.G. Verheijen
Teaching Staff	prof. dr. V.M. Heine
Teaching method(s)	Study Group, Lecture

Course Objective

The course Neuron-Glia interactions aims to go beyond the neuron-only view on functioning of the brain. Glial cells are as numerous as neurons and have traditionally been viewed to support neuronal functions, however, research of the last 2 decades suggests that glial cells may do more than just supporting neuronal function. In fact, for the past few years it has become evident that the functioning of the brain can be understood only if the interaction of all cell types, i.e. neurons and glial cells, is understood. This is a new understanding that is only slowly becoming part of textbooks. The aim of this ‘Neuron-Glia interactions’ course is to go educate MSc students in the contemporary insights on cellular processes that underlie brain function.
At the end of this course you will be able to:

- explain the cellular characteristics of astrocytes, oligodendrocytes and microglia cells.
- explain the functional role of the different glia cell types in the central nervous system.
- describe the current view on the implication of glia cell in different brain disorders, e.g. psychiatric, mood and neurodegenerative disorders.
- analyze how landmark Neuroscience papers (in the Glia field) are generated and composed.
- construct a research proposal (on a Glia research project).

Course Content

This course builds on existing knowledge, obtained from BSc and early MSc education, on neurobiology at the molecular, cellular and physiological level. Lectures from glia research expert from different universities will provide MSc students the required basic insight on neuron-glia interactions, which forms a frame work for later parts in the course were we will focus on thorough analysis of landmark papers in the Glia field, followed by the design and writing of a research proposal on Neuron-Glia interactions in health or disease.

Additional Information Teaching Methods

Lectures, supervised workgroups and individual assignments, self-study.

Method of Assessment

The grading procedure will be as follows:

- Grade of the contribution to workgroups on paper analyses: 10%
- Grade of research project presentation: 10%
- Research proposal: 30%
- Final exam (lecture content): 50%

The grades for the Research proposal and Final exam cannot compensate for each other and should both be at least 5.5. The final grade needs to be at least 5.5 to pass. For students that do not pass, one re-exam will be available. Participation in all workgroups is obligatory. Attending the lectures is highly recommended.

Literature

Required literature (research papers) will be provided at the start of the course.

Additional Information Target Audience

This course is for 2nd year VU Master Neurosciences students. Students with another background, please first contact the coordinator.

Additional Information

There is a maximum number of students (20). This means that students other than the target group should first contact the coordinator.

Recommended background knowledge

This course builds on existing knowledge, obtained from BSc and early MSc education, on neurobiology at the molecular, cellular and physiological level.

Neurophilosophy and Ethics

Course Code	AM_1018
Credits	3
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	L.C. de Bruin
Examiner	L.C. de Bruin
Teaching Staff	L.C. de Bruin, dr. B. Drukarch
Teaching method(s)	Lecture

Course Objective

At the end of this course students are able to:

1. explain and analyze the main philosophical and ethical issues in relation to recent developments in neuroscience: the mind-body problem, neuroscientific explanation and modeling, free will and predictability, neuroethics and animal ethics
2. critically assess the role of metaphors in the translation of a neuroscientific findings to the clinic and/or the public
3. reflect on and to discuss in a conceptually sound and ethically appropriate way the impact of neuroscientific findings on broader societal issues.

Course Content

This course provides students with an introduction to the most important philosophical and ethical issues in neuroscientific research, and aims to raise their awareness of conceptual and ethical issues in the translation of neuroscientific insights to professional practices and the public. Topics range from the mind-body problem, neuroscientific explanation and modeling (in particular mechanistic explanation and network models), to animal ethics and the implications of deep brain stimulation for personal identity. Students work in groups to investigate the potential of new metaphors for the translation of scientific research and do a Moral Case Deliberation (MCD) on a selected neuroscientific/neurotechnological case-study.

Additional Information Teaching Methods

Interactive lectures, video, discussion of the literature, workshops.

Method of Assessment

(A): Assignment New metaphors for translating neuroscientific research (50% of the final mark; assessment of learning objectives 1, 2)

(B): Assignment Moral Case Deliberation (MCD) on a selected neuroscientific/neurotechnological case-study (50% of the final mark; assessment of learning objectives 1, 3)

Literature

Will be provided on Canvas before the beginning of the course.

Additional Information Target Audience

VU Master Neurosciences students.

Rhythms of the Brain

Course Code	AM_1003
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. K. Linkenkaer-Hansen
Examiner	prof. dr. K. Linkenkaer-Hansen
Teaching Staff	prof. dr. K. Linkenkaer-Hansen
Teaching method(s)	Study Group, Computer lab, Practical, Lecture

Course Objective

The course "Rhythms of the Brain" is focused on measuring, analyzing, and interpreting the functional role of neuronal oscillations in humans.

At the end of the course, the student should be able to:

1. Explain how the human brain generates scalp electroencephalographic (EEG) signals, both ongoing oscillations and event-related potentials (ERPs).
2. Acquire practical experience with EEG (i.e., measure EEG, perform quantitative and statistical analysis to conclude about the relation between brain activity and cognition/behavior and present the results on a poster).
3. Explain key concepts of complex-systems science that have gained acceptance in the cognitive and behavioral neurosciences.
4. Apply state-of-the-art complexity-analysis techniques to EEG data and perceptual/ behavioral time series, and
5. …understand how these techniques can be applied in fundamental science and applied medical fields, e.g., for clinical trials and personalized medicine.
6. Explain the advanced techniques that estimate brain sources from the EEG signals and outline the possibilities and limitations based on own experiences.
7. Explain the rationale of so-called "integrated biomarkers" based on machine learning, use specialized toolboxes to compute them, and critically reflect on the pros and cons of this approach to functionally assess the state of a human brain based on the rhythms that it generates.

Course Content

Understanding the complexity of the human brain and mind is one of the greatest scientific challenges of the 21st century. To address these challenges, researchers increasingly adopt theories and methods used to study complexity in other natural systems. In this course, we give you a solid conceptual understanding of "complexity" and tools to study the complexity of the human brain through quantitative analysis of the brain rhythms that it generates and the variability in cognitive and behavioral tasks.
We consider it critical that students gain an in-depth understanding of the analytical tools in order to properly use and interpret the outcome of the different analysis techniques. This is achieved by covering the theory in the lectures followed by tutorials in the computer rooms. The concepts of "critical dynamics" and power-law scaling

behavior are carefully explained in the context of time-series analysis tools, generating mechanisms, and functional implications. Key concepts of complex networks and analytical tools to characterize them based on M/EEG data are also covered.
Another important component of the course is to teach you how to perform high-density EEG recordings of spontaneous brain activity during resting-state conditions and cognitive tasks and to analyze these signals with classical as well as modern complexity algorithms. You will work in small groups to record, analyze and present both data on EEG and its cognitive/behavioral correlates at the end of the course.
Finally, the importance of non-stimulus driven brain activity and cognition for brain-related disorders such as depression, dementia, insomnia or attention deficit and hyperarousal disorder is discussed in the context of normal variation in biomarkers and the associated challenges in objective diagnosis, prognosis, and treatment selection. We explain how data mining and -classification techniques from artificial intelligence can be used to integrate information from multiple biomarker algorithms to increase the accuracy of clinically relevant functional assessments. While the course is focused on understanding variability in human cognition and behavior in health and disease, the concepts and tools equally apply to research on common animal models.

Additional Information Teaching Methods

The study credits amount to 168 hours of study, which are divided approximately as follows:

Activity **Hours of study**

Lectures (l) **20**
Self study (literature and lecture sheets) **38**
Practicals in EEG lab (Prac) **8**
Computer practicals and project assignment (A) **36**
Journal club (Pres) **8**
Poster preparation (A) **18**
Preparation for exams (poster and written) **40**

Total **168**

Method of Assessment

Analysis and making research poster (R, 15%)
Presentation of research poster (Pres, 25%)
Written examination (E: 60%)

Compensation is not possible for any of these assessments.

Literature

- Nikulin VV, Linkenkaer-Hansen K, Nolte G, Lemm S, Müller KR, Ilmoniemi RJ, Curio G. A novel mechanism for evoked responses in the human brain. Eur J Neurosci. 2007;25:3146–54.
- Mazaheri A, Jensen O. Asymmetric amplitude modulations of brain oscillations generate slow evoked responses. J Neurosci 2008;28:7781–7.
- Jensen O, van Dijk H, Mazaheri A. Amplitude asymmetry as a mechanism for the generation of slow evoked responses. Clin Neurophysiol 2010.
- Nikulin VV, Linkenkaer-Hansen K, Nolte G, Curio G. Non-zero mean and asymmetry of neuronal oscillations have different implications for evoked responses. Clin Neurophysiol 2010;121:186–93.
- Hardstone R, Poil S-S, Schiavone G, Jansen R, Nikulin VV, Mansvelder HD, Linkenkaer-Hansen K. Detrended fluctuation analysis: A scale-free view on neuronal oscillations. Frontiers in Physiology. 3:450. doi:10.3389/fphys.2012.00450. 2012.
- Annotated sheets from lectures

Additional Information Target Audience

Masters and PhD students with interest in human brain function in general and EEG methodology in particular.

Additional Information

dr. K. Linkenkaer Hansen with guest lectures of dr. D.J.A. Smit

Statistics in Neurosciences

Course Code	AM_1276
Credits	6
Period	P1

Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S. van der Sluis
Examiner	dr. S. van der Sluis
Teaching Staff	dr. S. van der Sluis
Teaching method(s)	Study Group, Computer lab, Lecture

Course Objective

Students will

- acquire basic statistical knowledge of e.g. research design, distributions, measurement levels of data, and statistical tests.
- learn to formulate hypotheses, and select the most suited statistical analysis for a particular experiment or research design.
- learn to explore and test underlying assumptions, and formally report the results.
- learn how to perform statistical analyses in R.

Course Content

Statistical data analysis is the process of inspecting, cleaning, transforming, and modeling data in order to test scientific hypotheses and answer research questions. The lectures of this course will provide an overview of quantitative methods that are frequently used in neuroscience research. These include e.g. correlation, regression, (paired) t-test, (repeated measures) ANOVA, and multi-level analysis. We will also discuss concepts like p-values, the multiple testing problem, Type I and II errors, sampling, and statistical power. Each lecture will provide the theoretical background. The practicals and weekly obligatory assignments will guide students through a series of tailored research problems that they will tackle using the statistical package R. Students will receive hands-on experience in the main steps involved in statistical analyses: from the formulation of hypotheses, selection of the most appropriate test, checking of assumptions, cleaning of data, and running of analyses in R, to formally reporting the obtained results. This hands-on experience is invaluable for the internships in the first and second year of the Master of Neurosciences, and for your success as an independent researcher.

Additional Information Teaching Methods

Lectures, computer practicals, assignments, exam.

Method of Assessment

- weekly assignments (pass/fail: you need to pass all assignments before you can participate in the exam and passing all obligatory weekly assignments yields 1 point to the final score (i.e., 10%)).
- data analysis assignment (data portfolio; 30% of the total score)
- Written exam [60% of the total score: 20 MC questions (40% of the written exam) + 5 open ended questions (60% of the written exam)]

The final grade of Statistics in Neuroscience is based on the obligatory weekly assignments, a data analysis assignment, and a written exam. Statistics in Neuroscience is successfully completed if students handed in all weekly assignments and obtained grades of 5.5 or higher for both the exam and the data analysis assignment.

Entry Requirements

Students are assumed to be familiar with chapters 1-5, 7, and 9 of the book "Discovering statistics using R" by Field, Miles & Field before entering the course. The first lectures, practicals, and assignments will provide a short review of these chapters. A short diagnostic entry test is provided to give students insight into their knowledge of statistics at the start of the course.

Literature

The literature for the course consists of chapters from a book and several scientific papers.

Book:

- Andy Field Discovering Statistics using R, 1st edition (unless the 2nd edition is available by then!), Sage.
  - Chapters 1-7, 9, 10, 12, 13, 15 and 19 (19.1-19.6)

Additional reading, e.g.,

- Aarts et al 2014 Nat Neurosci; doi:10.1038/nn.3648
- Button et al. 2013 Nature Reviews Neuroscience; doi:10.1038/nrn3475
- Krzywinski & Altman, 2013 Nat Methods; doi:10.1038/nmeth.2738

Additional Information Target Audience

All students following the VU Master Neurosciences program are required to take this course.
A selective number of students from the MSc Biomedical Sciences can participate (ask your master coordinator).
Are you enrolled in another program and would you like to participate in this course? Send an e-mail to the course coordinator (s.vander.sluis@vu.nl) before the course starts so that your eligibility can be evaluated.

Additional Information

Coordinator: Sophie van der Sluis (s.vander.sluis@vu.nl)

Explanation Canvas

Canvas will be used to share files and upload assignments. Slack will be used for communication with and between students.

Systems Neuroscience

Course Code	AM_470712
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. T.J. de Vries
Examiner	prof. dr. T.J. de Vries
Teaching Staff	prof. dr. Y.D. van der Werf
Teaching method(s)	Seminar, Study Group

Course Objective

- Understanding of approaches to study the central nervous system in an integrated and multidisciplinary way with a strong focus on how the complexity of the brain is required for complex behaviour.
- The ability to write a research proposal from a systems neuroscience perspective.

Course Content

Systems Neuroscience is a "way of life": approaching the study of the central nervous system in an integrated and multidisciplinary way. Once learned in an exemplar system, the systems approach can be applied to essentially any functional system in the CNS. In this course we will explore the organization of essential systems, including those encoding for learning and memory, sleep, consciousness, decision making and reward. This will to a large extent consist of homework assignments with short presentations and discussion. Based on selected review- type papers/chapters we will a) follow the development of concepts over time; b) discuss the relationship between technology-development and experimental approaches c) study and discuss different approaches and d) integrate those into a concept of systems neuroscience.

Additional Information Teaching Methods

Lectures, homework assignments, presentations, and tutored discussions.

Contact hours: 24; Selfstudy: 48 hours.

Method of Assessment

Your final score for the course will depend on three items:

1. in class participation (33%)
2. writing a research proposal (group; 33%)
3. orally presenting a research proposal (group; 33%)

Literature

- Kandel ER, Schwartz JH, Jessell TM Principles of Neural Science. This book will be used in the course as background literature.
- Other literature will be provided during the course or will be self-selected.

Additional Information Target Audience

2nd year VU Master Neurosciences students.

Recommended background knowledge

Basic neuroscience.